

# Mate 6551

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## Tarea 4

4.7. Compute  $H_n(S^0)$  for all  $n \geq 0$ .

\*4.10. Prove that  $P_n$  is “natural”: if  $f: X \rightarrow Y$  is continuous, there is a commutative diagram

$$\begin{array}{ccc} S_n(X) & \xrightarrow{P_n^X} & S_{n+1}(X \times \mathbf{I}) \\ f_{\#} \downarrow & & \downarrow (f \times 1)_{\#} \\ S_n(Y) & \xrightarrow{P_n^Y} & S_{n+1}(Y \times \mathbf{I}). \end{array}$$

\*4.11. If  $X$  is a deformation retract of  $Y$ , then  $H_n(X) \cong H_n(Y)$  for all  $n \geq 0$ . In fact, if  $i: X \rightarrow Y$  is the inclusion, then  $H_n(i)$  is an isomorphism.

\*4.13. Prove that the Hurewicz map  $\varphi$  is “natural”. If  $h: (X, x_0) \rightarrow (Y, y_0)$  is a map of pointed spaces, then the following diagram commutes:

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{h_*} & \pi_1(Y, y_0) \\ \varphi \downarrow & & \downarrow \varphi \\ H_1(X) & \xrightarrow{h_*} & H_1(Y). \end{array}$$

\*4.15. Let  $X$  be a space and let  $\alpha, \beta, \gamma$  be (not necessarily closed) paths in  $X$  such that  $\alpha * \beta * \gamma$  is defined and is a closed path. Prove that, in  $H_1(X)$ ,

$$\text{cls}(\alpha * \beta * \gamma) = \text{cls}(\alpha + \beta + \gamma) = \text{cls } \alpha + \text{cls } \beta + \text{cls } \gamma.$$

(Hint: Use Theorem 4.27 and Exercise 3.4 with maps  $\Delta^2 \rightarrow X$  suggested by

